

COM-480: Data Visualization - Eurobnb Insights

Milestone 2 (18th April, 5pm)

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1. Project Goal

The aim of our interactive web-based platform is to visualize Airbnb listings in European cities; Amsterdam, Athens, Barcelona, Berlin, Budapest, Lisbon, London, Paris, Rome, and Vienna. This platform will allow users to explore rental trends, pricing, guest satisfactions and different location dynamics. Through dynamic filtering and a high quality of visualization of data, the platform provides an intuitive way to understand short-term rental patterns across major European cities.

2. Breakdown of our Goal and Visualization

Our primary objective is to uncover key trends and value differences in the short-term Airbnb rental markets in European cities. This goal is supported by two key analyses: City-to-City Comparisons and District-to-District Comparisons. These two approaches are crucial as they empower travelers to first identify which European city best fits their needs, and then determine the most suitable neighborhood within their chosen destination.

2.1. City vs. City Comparison

2.1.1 Parallel Coordinates:

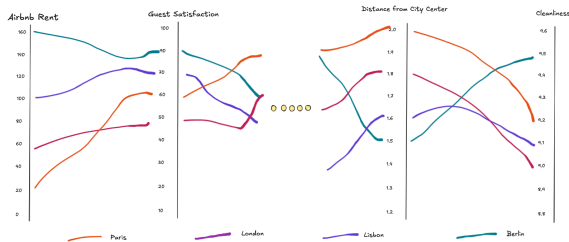


Figure 1: Parallel coordinates visualization comparing Airbnb listings across all key metrics

Parallel Coordinates Plots offer a clear way to compare Airbnb listings across metrics like price, satisfaction, and location, and handling hundreds of listings without losing readability. Figure 1 showcases how each listing is represented as a line across parallel axes, making trends, such as price-value trade-offs or regional differences instantly visible.

- **Technologies:** Plot.js, React, Python
- **Lectures:** Lecture 5, Lecture 2

2.1.2 Radar Chart (City Comparison of Key Metrics):

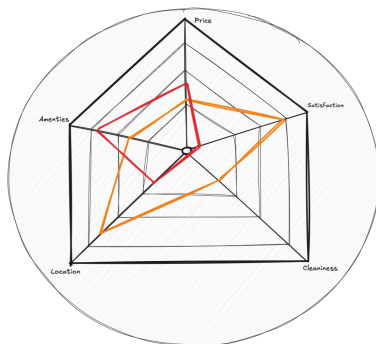


Figure 2: Radar chart comparing key attributes between City A and City B, including price, satisfaction, cleanliness, location, and amenities.

Radar Charts compares multiple metrics across cities in a single view in which it highlights strengths, weaknesses, and key patterns. The difference this visualization has as compared to other charts is the clear display of five or more variables like price, satisfaction and cleanliness. This would let user spot outliers and similarities at glance. As it can be seen on Figure 2, City Yellow surpasses City Red with better ratings and lower costs, while City Red struggles with cleanliness.

- **Technologies:** Chart.js, Victory, React, Python
- **Lectures:** Lecture 5 (Tabular Data), Lecture 2 (JavaScript)

2.2. District vs. District Comparison

2.2.1 Tiled Web Map with Dot Distribution/Clustering

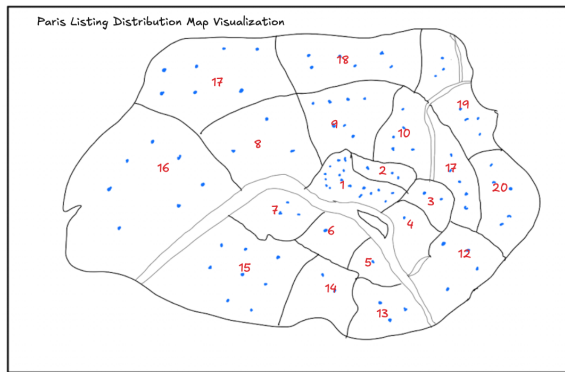


Figure 3: Mapping of all Listing in a City

This Interactive is important, as it would first showcase a city and visually show all the different Districts. Moreover, the interactive map would plot all Airbnb listings with district boundaries, using clustering to avoid overcrowding. Clicking any listing shows detailed information, while filters help narrow options by price, room type, or ratings.

- **Technologies:** Mapbox GL JS (react-map-gl), Leaflet, React, Python
- **Lectures:** Lecture 8 (Maps), Lecture 2 (JavaScript)

2.2.2 Weekend vs. Weekday Prices by District

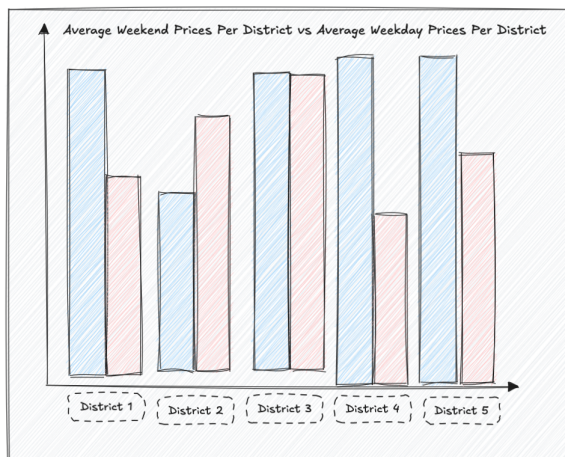


Figure 4: Sketch of Bar charts for Average Rental Price in Weekend Vs Average Rental Price in Weekdays

Figure 4 uses a comparative bar graph to show Airbnb price differences between districts on weekdays versus weekends, highlighting how accommodation costs shift temporally and geographically. This side-by-side visualization helps travelers spot neighborhoods with smaller weekend price hikes, enabling smarter budgeting for trips where lodging costs heavily influence decisions. More further Bar Graphs can be implemented depending on the results of more through Dataset Analysis.

- **Technologies:** Recharts, Nivo, React, Python
- **Lectures:** Lecture 11 (Tabular Data)

2.2.3 Interactive Treemap Vizulization

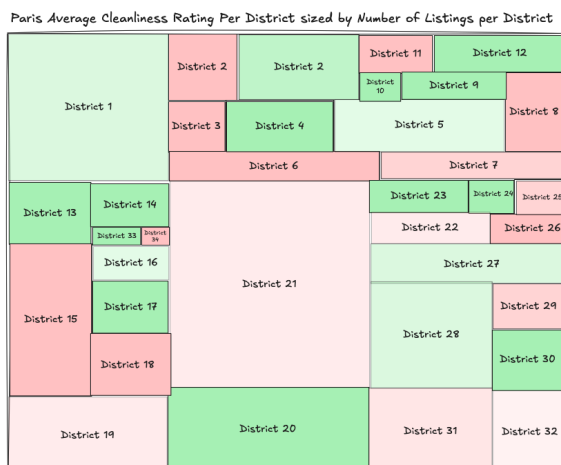


Figure 5: Interactive treemap visualizing district-level Airbnb data

This treemap makes it easy to spot patterns in Airbnb listings across different districts. The size of each area shows how many rentals are available—bigger squares mean more options—while the color can be switched to highlight things like price, cleanliness, or ratings. Want to know which busy areas also have the best deals? Just toggle the color to "price per rating" and the visual does the work for you.

- **Technologies:** D3.js, Nivo Treemap, React, Python
- **Lectures:** Lecture 10 (Graphs)

3. Extra Plausible Implementations

- **Smart Ranking System for Cities and Districts:** A dynamic scoring algorithm which can evaluate cities and districts based on key factors like Price, Guest Satisfaction, Location Convenience, and Amenities. If possible users can also get a preset personalized recommendations such as "Best budget-friendly district in Berlin".
- **Dynamic Bar Charts for Deeper Insights:** Instead of static chart, we can generate interactive bar graphs based on user-selected filters. These can be updated in real time as the users explore the data, making trends clearer.
- **Seamless Map-Centric Navigation:** Rather than jumping between separate widgets, all visualizations could integrate directly into map interface.

4. Implementation Breakdown

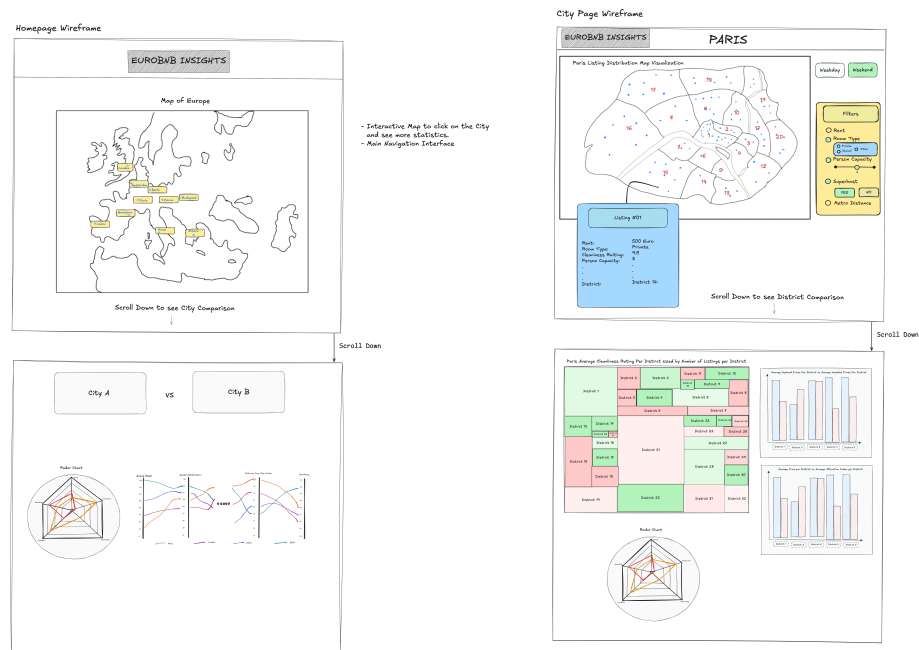


Figure 6: Wireframe Prototype of Eurobnb

Our Web-app would be implemented on React and be hosted on Github Pages. Through the react package called "gh-pages" we can use it to compile a "static" version of our website to be hosted on Github. As it can be seen on Figure 6, the homepage would have a map, showcasing all the European Cities as buttons which can lead to another page with more details for that city. Bellow the homepage navigation map, there would be a city-to-city analysis, where it showcases visualizations to help the viewer have better understanding of which city has better Aribnb listings. In the other hand the City Page would again have a map in which you can see all the listings in that city. Clicking on the listing would give statistics about that listing specifically. If you scroll down, you can see analysis on that city itself.

5. Prototype Website Review

Our prototype can be accessed at Airbnb Insights App Prototype. We have made a rough implementation of most of the widgets except of the visualizations one for city-to-city analysis. The Web App is build on ReactJS with Vite as frontend. In the Homepage there is a map with two labeled cities viewed as a button. This is a clickable button which would take to another page where the map of that city would be shown. Currently Amsterdam works as that has been implemented. There are rough implementations of a Treemap and Bart Chart implementations bellow that.

Repository Link: <https://github.com/com-480-data-visualization/eurobnb-insights>